AutoMergeNet: AutoML-Based M-Source Satellite Data Fusion Evaluated With Atmospheric Case Studies

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***Abstract*—Accurate detection of anomalous phenomena in satellite data often requires data layers containing complementary information (e.g., data from different sensors, auxiliary features, such as land cover maps, and metadata regarding data quality). However, existing highly specialized approaches to fuse multiple data layers cannot be transferred to other related problems, as they rely on expert-selected features and manual pipeline design. In this work, we propose *AutoMergeNet*, a framework for satellite image data fusion based on neural architecture search. AutoMergeNet generates neural networks that fuse any number of raster data layers. Consequently, it can address different classification problems based on satellite images without manual pipeline design. We designed the search space of AutoMergeNet by identifying relevant design choices from the image classification and data fusion literature. AutoMergeNet automatically transforms image classification networks into multibranch networks by optimizing critical architectural and training hyperparameters. Since the high dimensionality of multimodal image data poses a challenge for data fusion problems with limited labels, we use an auxiliary unimodal classifier combined with AutoMergeNet. We evaluate AutoMergeNet on a methane plume detection dataset from the literature and our newly created carbon monoxide plume detection dataset. AutoMergeNet performs strongly and consistently on these two multimodal classification problems, outperforming six baseline methods selected from state-of-the-art image classification approaches. Finally, we demonstrate the usability of our framework with a realistic methane plume detection use case, which shows that AutoMergeNet can be used as a highly specialized, state-of-the-art approach.**

***Index Terms*—Atmospheric plume detection, Earth observation (EO), multimodal image data fusion, neural architecture search (NAS).**

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I. Introduction

A COMMON problem in the Earth observation (EO) domain is the detection of a specific type of anomaly (e.g., oil spills on the surface of the sea or methane plumes in the atmosphere). This problem is often addressed by training a binary classifier, using a dataset where each image is labeled as either positive, showing "the anomaly of interest" or negative, depicting "anything else." This approach tends to misclassify phenomena in the satellite data that appear similar to the anomaly of interest. These phenomena, called "artefacts," lead to false positives through different mechanisms. For instance, algae on water can resemble oil spills in synthetic aperture radar (SAR) data, and variations in surface albedo can be erroneously interpreted as gas enhancements in the retrieval of gas concentrations from radiance spectra and be mistaken for methane plumes [1] (see Fig. 1).

Multimodal image data from different sources or measuring different physical quantities can reduce false positives. While both unimodal and multimodal image data can contain multiple channels (e.g., ImageNet's RGB channels [2]), they differ fundamentally. RGB channels are tightly linked, because they are obtained using a single sensor measuring one physical phenomenon—the number of photons collected per pixel. The channels inherit sensor biases, such as damaged pixels, and record information simultaneously. In contrast, multimodal image data are linked by time and location, but different sensors and physical phenomena provide multiple views of the subject with additional information and independent biases. Existing approaches show that multimodal image data can help reduce false positives by leveraging discerning features extracted from auxiliary data [1], [3], [4]. In methane plume detection, concentrations are inferred from satellite-measured spectra through "*retrieval.*" Data used or produced in the retrieval (such as albedo), meteorological data and outputs of physical models are subsequently used to reduce the number of false positives.